Training and Effort Dynamics in Apprenticeship

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Motivating Examples: Apprenticeships

- Work-for-training is common in a wide range of industries.
- Apprentices frequently experience long hours and heavy workloads.
  1. Initially a large share of their time is spent on menial work unrelated to the skills they wish to acquire.
  2. Gradually they progress toward more advanced tasks.

Apprenticeships in History

- Apprenticeships date back to at least the European trade guilds starting in the 12th century.
  1. They served as the main source of training for artisans and merchants.
  2. They gave rise to opportunities for exploitation.
- During the industrial revolution, long hours were commonplace and even a cause of public concern.
  1. Adam Smith considered industrial-revolution apprenticeships to be excessively long and poorly paid, as a response to the agent's liquidity constraints.
  2. A 14-hour workday was typical, with frequent cases of even longer hours.
  3. Apprenticeships back then commonly began with a period of menial work.

Restaurant Industry

- Star chefs possess coveted knowledge that aspiring chefs wish to acquire.
- Example from documentary *Jiro Dreams of Sushi*:
  1. Jiro, the world's top sushi chef, takes roughly ten years to train his apprentices.
  2. A large share of his apprentices' time is devoted to such monotonous tasks as cleaning and preparing seafood, including massaging octopus meat for 40 to 50 minutes per batch.

Science Careers

- In science careers, postdoctoral positions are widely used.
  1. Postdocs, especially in the life sciences, spend years working long hours at low wages in the hope of gaining skills and access to a tenure-track job.
  2. Many postdoc employers are able to train their postdocs very slowly, all the while using them as a form of cheap labor.
  3. "Many postdoctoral scholars—especially those not funded by training grants or fellowships—are but poorly paid research assistants who receive little mentoring and have few opportunities to develop an independent research agenda" (Stephan 2013).

Other Examples

- Surgery residencies:
  1. First-year trainees are assigned to "floor work" like "talking to patients' families".
  2. Over time, residents progress toward more complex procedures.
- Professional service firms:
  1. Young professionals seek to acquire knowledge from the firms' partners.
  2. Long hours and heavy workloads are common.
- Manicurists in the New York City area:
  1. They go through an apprenticeship reminiscent of those of the Industrial Revolution.
  2. As they acquire skills and pay their dues through menial work, they eventually advance through various career stages.

Main Results

This paper builds on Garicano and Rayo (2017).

- GR 2017 studies a dynamic self-enforcing contract in apprenticeships.
- In every profit-maximizing (principal-optimal) contract,
  1. the apprentice graduates in finite time,
  2. all available knowledge is transferred,
  3. the apprentice earns zero wages before graduation, and,
  4. the apprentice receives a knowledge gift in period 0.

Main Results

By endogenizing the effort provision, this paper shows that the optimal contract is inefficient in two channels:

- Apprentice works inefficiently hard (too hard).
- Apprentice spends excessive time on menial work.

Production, Payoffs, and Knowledge

- A principal/master (she) and an agent/apprentice (they).
- Over an infinite horizon in continuous time.
- Both players have discount rate r.
- At time t, the agent combines a stock of knowledge $X_t \in [0, \overline{X}]$ and two sorts of effort a_t, b_t to produce output y_t.
- Effort a_t is "skilled,"and its productivity is increasing in the agent's knowledge X_t.
- Effort b_t is "unskilled"or "menial,"and its productivity is independent of X_t.

Production, Payoffs, and Knowledge

- Total effort $q_t := a_t + b_t$.
- Both a_t and b_t are non-negative.
- q_t is bounded above by a constant normalized to 1.
- Total output $y_t := f(X_t, a_t) + g(b_t)$.
- The agent's flow payoff function is quasilinear in money.
- Exerting effort q_t imposes cost $c_t := c(q_t) \geq 0$ on the agent.

Production, Payoffs, and Knowledge

Assumption 1

(1) f and g are twice continuously differentiable with $f'_X > 0$, $f'_a \geq 0$, $f''_{Xa} > 0$, $f''_{aa} \leq 0$, $g' > 0$, $g'' \leq 0$, and either $f''_{aa} < 0$ or $g'' < 0$.

(2) c is twice continuously differentiable with $c' \geq 0$, $c'(0) = 0$, and $c'' > 0$.

- $v(X) := \max_{a,b \geq 0} \{f(X, a) + g(b) - c(a + b)\}$ denotes first-best surplus given X.
- $f(X, a)$ is strictly increasing in X, so is $v(X)$.

Assumption 2

$q^*(X) < 1$ for all $X \leq \overline{X}$.

Production, Payoffs, and Knowledge

- $y^*(X, q) := \max_{a \in [0,q]}\{f(X, a) + g(q - a)\}$ denotes the maximum possible output given knowledge and effort levels (X, q).
- $a^*(X, q) := \arg\max_{a \in [0,q]}\{f(X, a) + g(q - a)\}$ denotes the output-maximizing skilled effort given (X, q).
- $q^*(X) := \arg\max_{q \in [0,1]}\{y^*(X, q) - c(q)\}$ is the first-best level of total effort, which is unique from Assumption 1.
- $a^*(X) := a^*(X, q^*(X))$ is the corresponding first-best level of skilled effort, which is also unique.

Production, Payoffs, and Knowledge

- The agent starts with some exogenous stock of knowledge $\underline{X} \in [0, \overline{X}]$.
- Their stock of knowledge can never decrease, and the only way it can increase is by transfers from the principal.
- The principal is able to costlessly and instantaneously increase the agent's knowledge to any level up to $\overline{X}$.
- She can select any weakly increasing function X_t with range in $[\underline{X}, \overline{X}]$.

Apprenticeship Contracts

- The principal has full commitment power, and offers a contract to the agent at time 0.
- The contract specifies a graduation date $T \geq 0$ and for each $t \in [0, T]$ a knowledge stock X_t and effort levels (a_t, b_t).
- Between time 0 and T, all output belongs to the principal. After T, the agent works on their own and keeps all output.
- The agent can walk away at any time; if the agent does so, the principal does not hire them back.

Apprenticeship Contracts

- Consider only contracts with a finite graduation date T, and $X_T = \overline{X}$.
- The agent earns zero wages, and total effort $q_t = a_t + b_t$ at each time is allocated to maximize output, so that $y_t = y^*(X_t, q_t)$.
- The set of feasible contracts Ψ is the set of all graduation times T and time paths of effort and knowledge $(q_t, X_t)_{t=0}^T$ with
  1. $q_t : [0, T] \to [0, 1]$ satisfying the Lipschitz condition, that is, there exists a constant K such that $|q_t - q_s| \leq K|t - s|$ for all $t, s \in [0, T]$.
  2. $X_t : [0, T] \to [\underline{X}, \overline{X}]$ non-decreasing and right continuous.
- $\mathbf{q} := (q_t)_{t=0}^T$ and $\mathbf{X} := (X_t)_{t=0}^T$ denote the time paths of effort and knowledge, and $\mathcal{C} := (T, \mathbf{q}, \mathbf{X})$ denotes a contract.

Apprenticeship Contracts

- The agent has access to the same output technology when working for the principal and when working on their own. (general human capital)
- The agent also has access to an alternate employment that pays $\underline{v}$, with $0 < \underline{v} < v(\overline{X})$, so that total surplus is maximized by fully training the agent.
- After graduation the agent's flow payoff is $v(\overline{X})$.
- If the agent walks away before graduation with knowledge X, they obtain instantaneous surplus $\max\{\underline{v}, v(X)\}$, which will then be their flow payoff.

Apprenticeship Contracts

- Given a contract $\mathcal{C} \in \Psi$, the principal's and agent's continuation values from date $t \leq T$ onward are

$$\Pi_t(\mathcal{C}) = \int_t^T \mathrm{e}^{-r(\tau - t)} y_\tau \mathrm{d}\tau$$

and

$$U_t(\mathcal{C}) = \underbrace{\mathrm{e}^{-r(T-t)} \frac{1}{r} v(\overline{X})}_{\text{discounted "graduation prize"}} - \underbrace{\int_t^T \mathrm{e}^{-r(\tau - t)} c(q_\tau) \mathrm{d}\tau}_{\text{"effort tax"}} .$$

- "graduation prize" from date $T = \int_T^{+\infty} \mathrm{e}^{-r(\tau - T)} v(\overline{X}) \mathrm{d}\tau = \frac{1}{r} v(\overline{X})$

Apprenticeship Contracts

- The principal can select any contract she desires subject to the agent's *participation constraint*: for all $t \leq T$, the agent's continuation value should be at least as high as their outside option.
- The principal wishes to raise X_t to the point where PC holds with equality, because training is costless and increases the agent's productivity.
- Once the agent accepts the contract and the principal has the chance to raise the agent's knowledge to the desired level X_0, PC holds with equality from then on.
- Since the principal will immediately increase the agent's knowledge to at least the point where $v(X) = \underline{v}$, we write the agent's initial outside option as $(1/r)v(\underline{X})$, instead of $(1/r)\max\{\underline{v}, v(X)\}$.

Apprenticeship Contracts

- The participation constraint reduces to

$$U_0(\mathcal{C}) \geq \frac{1}{r} v(\underline{X}), \qquad U_t(\mathcal{C}) \geq \frac{1}{r} v(X_t) \text{ for all } t \in [0, T]. \tag{A-PC}$$

We call the first of these conditions the *initial participation constraint*, and the second one the *ongoing participation constraint*.

- Rewriting the (binding) ongoing participation constraint in differential form,

$$\mathrm{e}^{-r(T-t)} v(\overline{X}) + c(q_t) - r \int_t^T \mathrm{e}^{-r(\tau-t)} c(q_\tau) \mathrm{d}\tau = \frac{1}{r} \mathrm{d}v(X_t)/\mathrm{d}t,$$

$$\mathrm{e}^{-r(T-t)} \frac{1}{r} v(\overline{X}) - \int_t^T \mathrm{e}^{-r(\tau-t)} c(q_\tau) \mathrm{d}\tau = \frac{1}{r} v(X_t)$$

Apprenticeship Contracts

- It follows that

$$\underbrace{\mathrm{d}v(X_t)/\mathrm{d}t}_{\text{flow value of new knowledge}} = \underbrace{r[v(X_t) + c(q_t)]}_{\text{agent's opportunity cost}} .$$

  When $c_t = 0$, then $v(X_t)$ needs to increase at rate r.

- The principal needs to transfer knowledge more quickly when the agent's knowledge stock and outside opportunities are higher.
- The principal needs to transfer knowledge even more quickly the more she asks the agent to work.

Apprenticeship Contracts

- Formally, the principal's problem is

$$\max_{\mathcal{C}\in\Psi} \int_0^T \mathrm{e}^{-rt} y^*(X_t, q_t)\mathrm{d}t, \quad \text{s.t. A-PC.} \tag{3}$$

- Any optimal contract is sequentially optimal, so while the principal has commitment power, she does not make commitments that she would later prefer to undo.
- This is because the agent's ongoing participation constraint is always binding, so any feasible change the principal would want to make at some time t could not make the agent worse off from then on.
- Any potential improvement to the contract could be implemented ex ante.

Knowledge Premium

- Define the *knowledge premium* given knowledge X and skilled effort a as
$$\rho(X, a) := \frac{f'_X(X, a)}{f'_X(X, a^*(X))}.$$
- It follows from the envelope theorem that $f'_X(X, a^*(X)) = v'(X)$ (recall that $v(X) := \max_{a,b\geq 0}\{f(X, a) + g(b) - c(a + b)\}$),
- This premium measures the marginal impact of knowledge on the agent's productivity inside the relationship relative to its impact on the agent's outside option.
- For any given a, an increase in X that raises the agent's outside option $v(X)$ by 1, raises current output by $\rho(X, a)$.

Relaxed Problem

- Begin by solving a relaxed version of problem (3), where the principal is allowed to select the initial knowledge level X_0 to be arbitrarily low.
- The principal's problem simplifies to that of selecting any T and $\mathbf{q}$ while setting $\mathbf{X}$ to its highest level such that A-PC is met. That is,

$$v(X_t) = \mathrm{e}^{-r(T-t)} v(\overline{X}) - r \int_t^T \mathrm{e}^{-r} c(q_\tau) \mathrm{d}\tau. \tag{4}$$

- Let $\rho_t := \rho(X_t, a_t)$ denote the knowledge premium at time t, where $a_t = a^*(X_t, q_t)$.

Relaxed Problem

Theorem 1

In the relaxed problem the optimal contract is unique and satisfies

(1) $r\int_0^T \rho_t \mathrm{d}t = 1.$

(2) For all t,

$$\frac{\partial y^*}{\partial q}(X_t, q_t)/c'(q_t) = \max\left\{r\int_0^r \rho_\tau \mathrm{d}\tau, \frac{\partial y^*}{\partial q}(X_t, 1)/c'(1)\right\}. \tag{5}$$

As a result, the optimal effort path is efficient at the terminal time T, and exceeds the efficient level at all earlier times.

Optimal Graduation Time

INTUITION

- Consider first the choice of T. The principal faces a trade-off between a longer apprenticeship and a less-productive apprentice.
- Suppose that she lengthens the apprenticeship to $T + \mathrm{d}T$, while specifying the same effort as before up to T, and first-best effort over the interval $\mathrm{d}T$.
  1. This modification gains the principal, and costs the agent, 1 in present value.
  2. To meet PC, at each time $t \leq T$ the agent's outside option must fall by 1 in present value, which lowers their productivity by $r\rho_t$ in present value; hence, the principal loses $r\int_0^T \rho_t \mathrm{d}t$.
- It is therefore optimal to set $r\int_0^T \rho_t \mathrm{d}t = 1$.
- Because $\rho_t \geq 1$, the contract will last at most $1/r$ years.

Optimal Effort Path

INTUITION

- Consider the choice of effort path $\mathbf{q}$. The principal faces a trade-off between a higher output today and lower output at every prior date.
  1. Suppose that she raises effort at time t so as to cost the agent 1 extra in present value.
  2. Recall that
  $$U_t(\mathcal{C}) = \mathrm{e}^{-r(T-t)}\frac{1}{r}v(\overline{X}) - \int_t^T \mathrm{e}^{-r(\tau-t)}c(q_\tau)\mathrm{d}\tau.$$
  3. This raises output by $\frac{\partial y^*}{\partial q}(X_t, q_t)/c'(q_t)$ in present value
  4. To meet PC, at each time $\tau \leq t$ the agent's outside option must fall by 1 in present value, which lowers their productivity by $r\rho_t$ in present value; and so the principal's overall loss is $r\int_0^r \rho_\tau \mathrm{d}\tau$.

Optimal Effort Path

INTUITION

- The principal either sets

$$\frac{\partial y^*}{\partial q}(X_t, q_t)/c'(q_t) = r\int_0^r \rho_\tau \mathrm{d}\tau \tag{6}$$

  or, when this is impossible given the upper bound on effort, she sets $q_t = 1$.
- The RHS of (6) is the principal's (shadow) cost of effort. The LHS of (6) is the (inverse) effort distortion.
- This ratio is strictly decreasing in effort, and is equal to 1 when effort is first-best.
- Since effort is bounded above by 1, the lowest feasible value for the ratio is $\frac{\partial y^*}{\partial q}(X_t, 1)/c'(1)$.

Optimal Effort Path

- Early in the relationship the (shadow) cost of effort is so low that the principal optimally sets $q_t = 1$.
- As time goes by this cost grows, and effort becomes less and less distorted relative to first best; until, at the terminal time, the principal fully internalizes the agent's true cost and effort becomes first best.

Uniqueness

INTUITION

The two equations in the theorem, together with the binding A-PC (4), define a unique contract.

- Begin at time T where knowledge is known to be $\overline{X}$, effort is first best, and $\rho_T = 1$.
- Move backward in time using the system (4) and (5), the relationship $r\int_0^t \rho_\tau \mathrm{d}\tau = 1 - r\int_t^T \rho_\tau \mathrm{d}\tau$, and $\rho_t := \rho(X_t, a_t)$ to determine the time-paths of knowledge, effort, and knowledge premia.
- Because (4) and (5) uniquely define X_t and $\mathbf{q}_t$ given the values of $\int_t^T \mathrm{e}^{-r} c(q_\tau)\mathrm{d}\tau$ and $\int_t^T \rho_\tau \mathrm{d}\tau$, these paths are unique.

Original Problem

Theorem 2

In the original problem the optimal contract is unique. Moreover, for all $\overline{X}$, there is a threshold $\hat{X} < \overline{X}$ such that

(1) if $\underline{X} \leq \hat{X}$, the optimal contract corresponds to the relaxed solution;

(2) if $\underline{X} > \hat{X}$, the optimal contract is the truncated version of the relaxed solution obtained by eliminating from this relaxed solution all time periods where knowledge is below $\underline{X}$. That is, the agent begins the contract farther along, with initial knowledge $X_0 = \underline{X}$.

- This threshold solves $v(\hat{X})/r = U_0$, where U_0 denotes the agent's total payoff in the relaxed solution.

Original Problem

- The first part of the theorem is immediate.
- For the second part, notice that
  1. The agent's ongoing participation constraint always binds, the so unconstrained solution is sequentially optimal.
  2. Whenever $\underline{X} > \hat{X}$, it is optimal to begin the contract with knowledge $\underline{X}$ and to proceed from there as prescribed by the unconstrained solution.
- When $\underline{X} < \hat{X}$ the principal grants the agent a knowledge gift at time 0, raising the agent's knowledge to level $\hat{X}$.
- An example would be a summer crash course at the beginning of a consulting career.

Pareto-Efficient Contracts

- The principal's optimal contract is sequentially optimal. As time goes by, the players' continuation payoffs trace the Pareto frontier.
- Any Pareto-efficient payoffs (Π, U) are uniquely implemented by an initial knowledge gift to raise the agent's knowledge to $v^{-1}(rU)$.
- Every Pareto-efficient contract is a truncated version of the principal's optimal contract, where the agent begins the contract farther along and therefore endures a shorter apprenticeship.

Pareto-Efficient Contracts

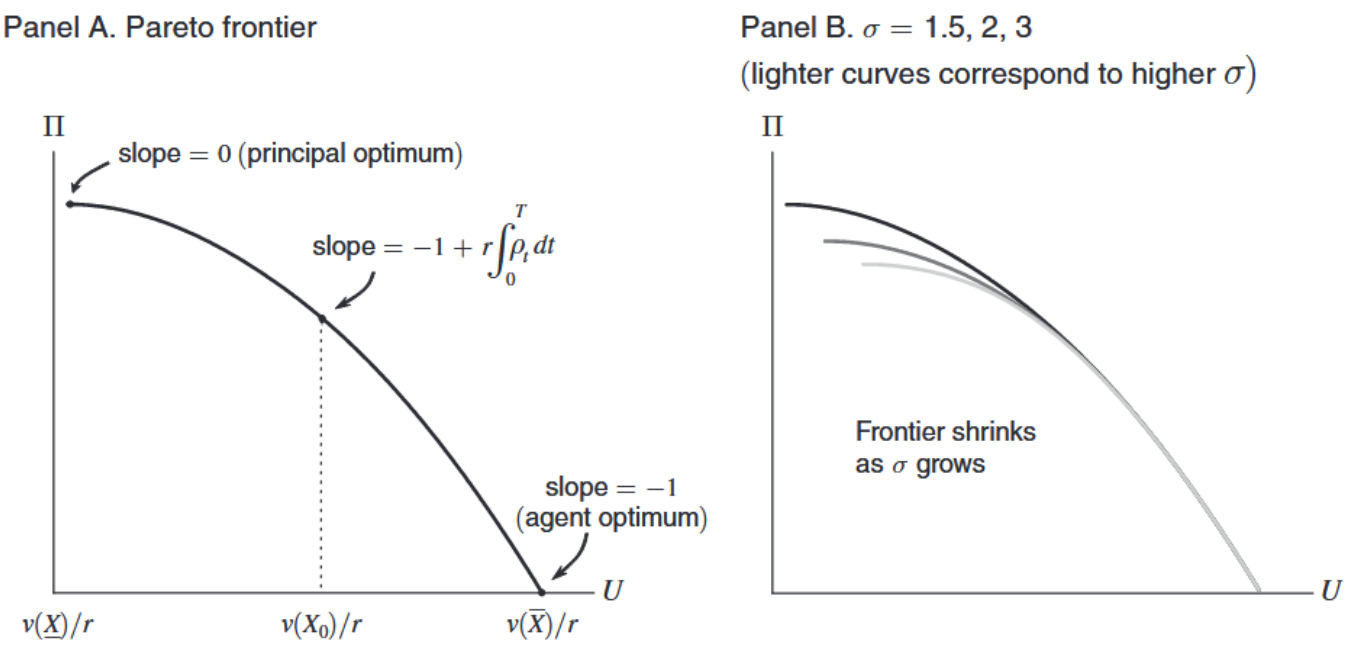

FIGURE 1

- Figure 1 depicts the Pareto frontier for the case where $y^*(X, q) = X + q$ and $c(q) = \alpha + \beta q^\sigma$.

Evolution of Efforts

Corollary 1

The optimal path of unskilled effort $\mathbf{b} = (b_t)_{t=1}^{T}$ *is weakly decreasing, and it is strictly decreasing whenever unskilled effort is positive and below its upper bound of* 1.

- To gain intuition, notice that two effects cause b_t to fall.
  1. As time goes by the principal's shadow cost of effort grows, so unskilled effort becomes less distorted relative its first-best level.
  2. The marginal value of skilled effort grows, so the first-best level of unskilled effort falls.

Contract Length

Corollary 2

When the agent's initial participation constraint is slack and the contract specifies a first-best level of skilled effort at all times, then $T = 1/r$; otherwise $T < 1/r$.

- If skilled effort is first-best at all times, $\rho_t \equiv 1$.
- The optimal length T satisfies $r\int_0^T \mathrm{d}t = rT = 1$, which corresponds to the optimal length in GR 2017.

Extensions

- Training Certificates (and Indentured Servitude): incorporating graduation certificate and penalty.
- Cash Payments: allowing agent to start with cash balance M.
- Caps on Effort and Apprenticeship Length

Caps on Effort and Apprenticeship Length

Proposition 1

For every Pareto-efficient contract $\mathcal{C}$ there is a cap on contract length and a (time-dependent) cap on effort that together induce the principal to select $\mathcal{C}$.

Caps on Effort and Apprenticeship Length

Proposition 2

Let $(T, \mathbf{q}, \mathbf{X})$ *denote the principal-optimal contract and suppose* $q_t \geqslant q^*(\overline{X})$ *for all* t*. Then any cap on contract length* $T^{CAP} \leqslant T$ *combined with a constant cap on effort* $q^{CAP} = q^*(\overline{X})$ *weakly increases surplus, strictly so whenever* $T^{CAP} < T$*, or* $q_t > q^*(\overline{X})$ *at some time* t*, or both.*

Caps on Effort and Apprenticeship Length

- But without the hypothesis in Proposition 2, even very mild caps on length or effort alone can be counterproductive.

Remark 1

Let $(T, \mathbf{q}, \mathbf{X})$ *denote the principal-optimal contract absent regulation. Suppose knowledge and effort are substitutes (i.e.,* $y^*(X, q) = X + q$*) and suppose* $\underline{X}$ *is strictly above the unconstrained initial knowledge level, and so there is no initial knowledge gift. Then, every cap on contract length* $T^{CAP} < T$ *that is close to* T *reduces surplus. Similarly, every cap on effort* $q^{CAP} < \sup_t q_t$ *that is close to* $\sup_t q_t$ *reduces surplus.*

Conclusion

- This paper studies the optimal contract for a principal with commitment power to "sell" knowledge to a cash-constrained agent who is free to walk away at any time.
- The optimal contract is inefficient both because (1) the agent will work inefficiently hard to compensate the principal for their training, and (2) this training is inefficiently slow.
- Bounds on effort can lead to even slower training, and bounds on apprenticeship length can increase overwork, but total surplus can be increased by a cap on apprenticeship length combined with a judiciously chosen upper bound on effort.
- These results follow from the assumption that the agent is unable to commit to keep working for the principal after being trained.